AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 35 · STUDENT LESSON

Capstone limitations and revision

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

After selecting rate 0.2, may you change it after viewing final scores and keep calling those scores untouched?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Evaluate a frozen candidate and baseline on final rows. Try the recall before opening notes.

Your learning target

I can evaluate a frozen candidate and baseline on final rows.

I can identify a distribution limitation from raw errors.

I can distinguish **revision** checks from new independent evaluation.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone controlled experiments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Documenting a network experiment. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Uncertainty and failure analysis. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Evaluate a frozen candidate and baseline on final rows

Identify a distribution limitation from raw errors

Distinguish **revision** checks from new independent evaluation

Retrieve what you already know

> After selecting rate 0.2, may you change it after viewing final scores and keep calling those scores untouched?

My starting explanation

This week's end goal

Explain capstone limitations and **revision** with a traceable toy-network case and a bounded conclusion.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Capstone limitations and revision case. Separate observations, given facts and assumptions before choosing an explanation.

Freeze the Week 34 candidate: max(0, a*x), initial a = 1, rate 0.2, two full-batch updates on (1,2), (2,4), no bias. The final weight is 2. The baseline is unchanged at a = 1.

Now open final rows F1=(5,10), F2=(6,12), F3=(-1,-2), F4=(-2,-4). These fictional rows deliberately include a condition absent from the positive **training** and validation **input**s.

Opening them is an evaluation event, not permission to tune on them while retaining an independent-test claim.

Write raw **prediction**s for both procedures before summarising. A low overall error can hide a systematic problem in a condition. These four constructed examples do not estimate prevalence or real-world performance.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Capstone limitations and revision. Connect each labelled input, deciding step and result.

ID · x · y · Baseline prediction · Candidate prediction

> **ID:** F1
>
> **x:** 5
>
> **y:** 10
>
> **Baseline prediction:** 5
>
> **Candidate prediction:** 10

> **ID:** F2
>
> **x:** 6
>
> **y:** 12
>
> **Baseline prediction:** 6
>
> **Candidate prediction:** 12

> **ID:** F3
>
> **x:** -1
>
> **y:** -2
>
> **Baseline prediction:** 0
>
> **Candidate prediction:** 0

> **ID:** F4
>
> **x:** -2
>
> **y:** -4
>
> **Baseline prediction:** 0
>
> **Candidate prediction:** 0

Baseline squared errors are 25,36,4,16, so MSE is 81/4 = 20.25. Candidate squared errors are 0,0,4,16, so MSE is 20/4 = 5. The candidate passes the strict lower-MSE final criterion on these four rows.

On the two negative-**input** rows its MSE is (4+16)/2 = 10; on the positive rows it is zero.

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Frozen candidate a = 2	Final MSE = 5	Improves on baseline 20.25
Negative x cases	Predictions forced to zero	A structural limitation
Change output rule	Check revealed failures	New final evidence needed

A passed aggregate criterion and a known structural failure can both be true.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Capstone limitations and revision. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Compute both overall MSE values and the candidate negative-**input** MSE. Apply the strict final criterion.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Train the **revision** on all eight failed-slice evaluation patterns and rescore them.

Predicted result

Prediction: Treat the score as exposed-case development, not independent **evidence** of shift robustness.

Why the result follows

Reason: The evaluation records now influenced learning and model selection.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Capstone limitations and revision to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Frozen procedure	A recorded procedure fixed before final evaluation
Residual	Prediction minus observed target
Revision	A documented change informed by earlier evidence
Coverage limitation	A condition or group missing or poorly represented in the evidence

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

Follow the worked trace for Capstone limitations and revision, predict a step, then compare with its explanation.

Worked explanation · Capstone limitations and revision

The ReLU **output** cannot be negative. When targets are negative, no choice of this scalar weight can produce the required negative **output**. More epochs do not remove that **output** constraint.

A proposed signed linear **output** a*x would represent negative targets, but it changes the architecture. It needs a new version, appropriate **fitting** and selection, and fresh reserved evaluation.

Worked explanation · Capstone limitations and revision

Running that proposal on F3 and F4 checks the revealed failures. It does not create new independent final **evidence**. Preserve the original test result and label later scores as **revision** or regression checks.

Do not replace the earlier failure report with a score from a new model on already inspected rows.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone limitations and revision

Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim.

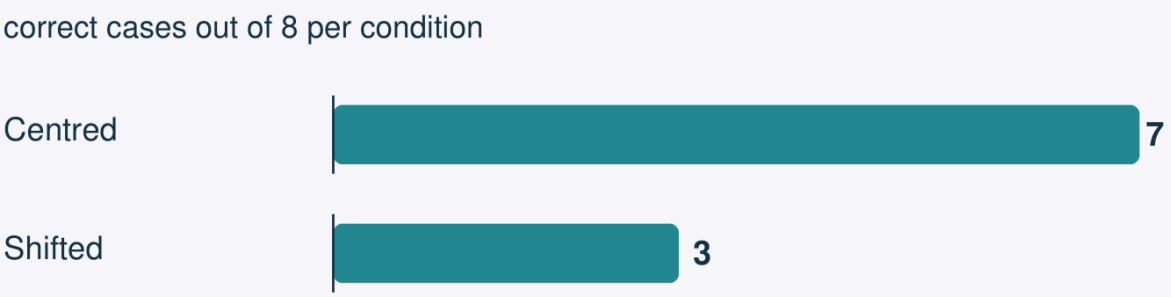
The facts we will use

Supplied facts

Centred slice: 7/8 correct. Shifted slice: 3/8 correct. The network and label rubric are frozen.

AI ACADEMY / AI-5/35

Toy network condition slices



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/35

Capstone limitations and revision

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute slice rates

Centred is 87.5%; shifted is 37.5%.



2. Revise the claim

Do not describe the network as shift-robust on this evidence.



3. Plan a fair revision

Use separate shifted development patterns, then assess the frozen revision on reserved shifted cases.

CHANGE ONE THING

Train the revision on all eight failed-slice evaluation patterns and rescore them.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute slice rates	Centred is 87.5%; shifted is 37.5%.
2. Revise the claim	Do not describe the network as shift-robust on this evidence.
3. Plan a fair revision	Use separate shifted development patterns, then assess the frozen revision on reserved shifted cases.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A limitation guides a better next experiment. **Training** on a failed evaluation example may help development, but that exposed example cannot remain the untouched **evidence** for the **revision**.

What this example does not establish

Keep the stated limit

Small slices may vary; this example diagnoses an observed weakness without inventing a population rate or cause.

Keep the stated limit

Read the labelled picture: Calculate the slice rates and explain the claim **revision**. If the team trains on the shifted evaluation patterns, what kind of **evidence** does their rerun provide?

Keep the stated limit

Change one thing: Train the **revision** on all eight failed-slice evaluation patterns and rescore them. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Capstone limitations and revision claim and explain a repair using the supplied evidence.

Check the claim against the facts

Supported example: for a = 2 on x = -3 with target -6, the ReLU **prediction** is 0, **residual** 6 and squared error 36. The signed proposal predicts -6 on this revealed case.

This demonstrates that the alternative can represent the case; it does not establish how well a newly trained model will perform elsewhere.

Guided: calculate both predictions and errors and explain which claim the comparison supports.

Locate the earliest unsupported step

Why cannot extra **training** alone make this ReLU **output** negative? Name a proposed change and the **evidence** it would require.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Capstone limitations and revision with support, then explain the deciding step in your own words.

For F1–F4 calculate candidate **residual**s and squared errors.

Guided: calculate both predictions and errors and explain which claim the comparison supports.

Explain your prediction to a partner before comparing results.

Practise with less help: Compare with the baseline

Use workbook W2 for the connected example “Capstone limitations and **revision**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone limitations and **revision**. Proposed explanation: “Passing a criterion means no failures remain”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Calculate the slice rates and explain the claim **revision**. If the team trains on the shifted evaluation patterns, what kind of **evidence** does their rerun provide?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Capstone limitations and revision. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Compute both overall MSE values and the candidate negative-**input** MSE. Apply the strict final criterion.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Capstone limitations and revision. Explain why each step is justified.

Explain → apply → challenge

For F1–F4 calculate candidate **residual**s and squared errors.

Why cannot extra **training** alone make this ReLU **output** negative? Name a proposed change and the **evidence** it would require.

After inspecting F1–F4, a student changes to **prediction** 2*x and reports “new final MSE 0”. Explain the accurate label and the next evaluation step.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: A frozen ReLU candidate with a = 3 is compared with baseline a = 1 on final rows (2,6),(4,12),(-1,-3),(-2,-6).

Compute both MSE values and candidate MSE on negative **input**s. Write a bounded conclusion. Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

A **revision** record names the observed failure, proposed **mechanism**, changed rule, preserved earlier results and new evaluation plan. Reserve fresh cases spanning the relevant **input** conditions.

Include regression checks for earlier successes, and keep independent evaluation separate. Record any cases still missing from coverage.

> **Exit: write one sentence reporting the original final result and one sentence describing what remains unverified after the proposed architecture change.**

After inspecting F1–F4, a student changes to **prediction** 2*x and reports “new final MSE 0”. Explain the accurate label and the next evaluation step.

> **My limited claim and a useful next check**

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Capstone limitations and revision case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
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Negative x cases	Predictions forced to zero	A structural limitation
Change output rule	Check revealed failures	New final evidence needed

A passed aggregate criterion and a known structural failure can both be true.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Evaluate new evidence independently

A frozen ReLU candidate with a = 3 is compared with baseline a = 1 on final rows (2,6),(4,12), (-1,-3),(-2,-6). Compute both MSE values and candidate MSE on negative **input**s. Write a bounded conclusion.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A frozen ReLU candidate with a = 3 is compared with baseline a = 1 on final rows (2,6),(4,12), (-1,-3),(-2,-6). Compute both MSE values and candidate MSE on negative **input**s. Write a bounded conclusion.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Capstone limitations and revision, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim.

By the end, I can	How I will show it
Evaluate a frozen candidate and baseline on final rows	Explain the deciding step, show a trace or test, and state one limit.
Identify a distribution limitation from raw errors	Explain the deciding step, show a trace or test, and state one limit.
Distinguish revision checks from new independent evaluation	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Capstone limitations and revision without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Evaluate a frozen candidate and baseline on final rows

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases.

1. Compute slice rates

2. Revise the claim

3. Plan a fair **revision**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Plan a fair **revision** Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases.

A limitation guides a better next experiment. **Training** on a failed evaluation example may help development, but that exposed example cannot remain the untouched **evidence** for the **revision**.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Train the **revision** on all eight failed-slice evaluation patterns and rescore them. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Treat the score as exposed-case development, not independent **evidence** of shift robustness.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The

answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Treat the score as exposed-case development, not independent **evidence** of shift robustness. The evaluation records now influenced learning and model selection.

Repair a missing building block

Review Capstone controlled experiments (ai-5/34). Goal: Trace matched two-update learning-rate experiments. Short practice: Compute the two correct counts and identify the changed case.

Why would changing B's **training** data weaken the conclusion about its added connection?. Return to this week and explain what you changed.

Review Documenting a network experiment (ai-5/32). Goal: Record a network experiment with enough detail to replay its calculations. Short practice: What additional **evidence** is needed for a final performance claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Uncertainty and failure analysis (ai-5/29). Goal: Distinguish a predicted class, confidence score and observed correctness. Short practice: What is C's positive-class probability and chosen-label confidence?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

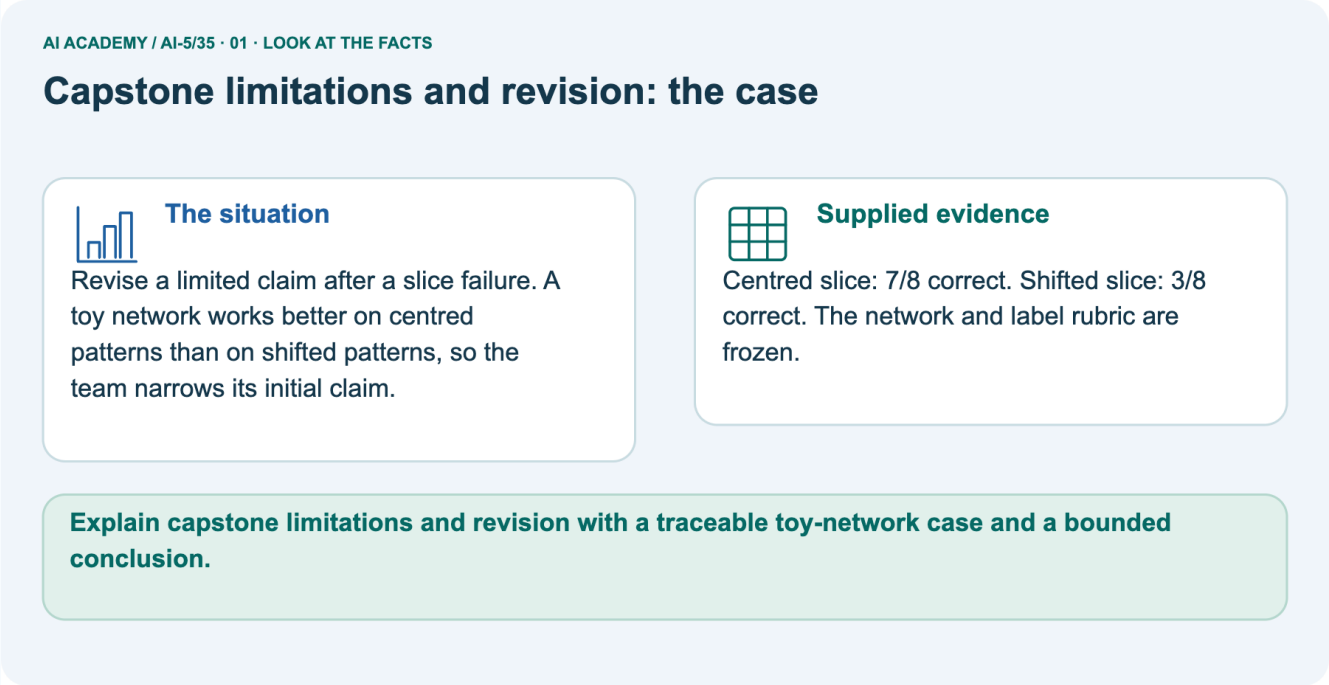
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Capstone limitations and revision graphic using its labels, deciding evidence and relevant limitation.

01. Capstone limitations and revision: the case



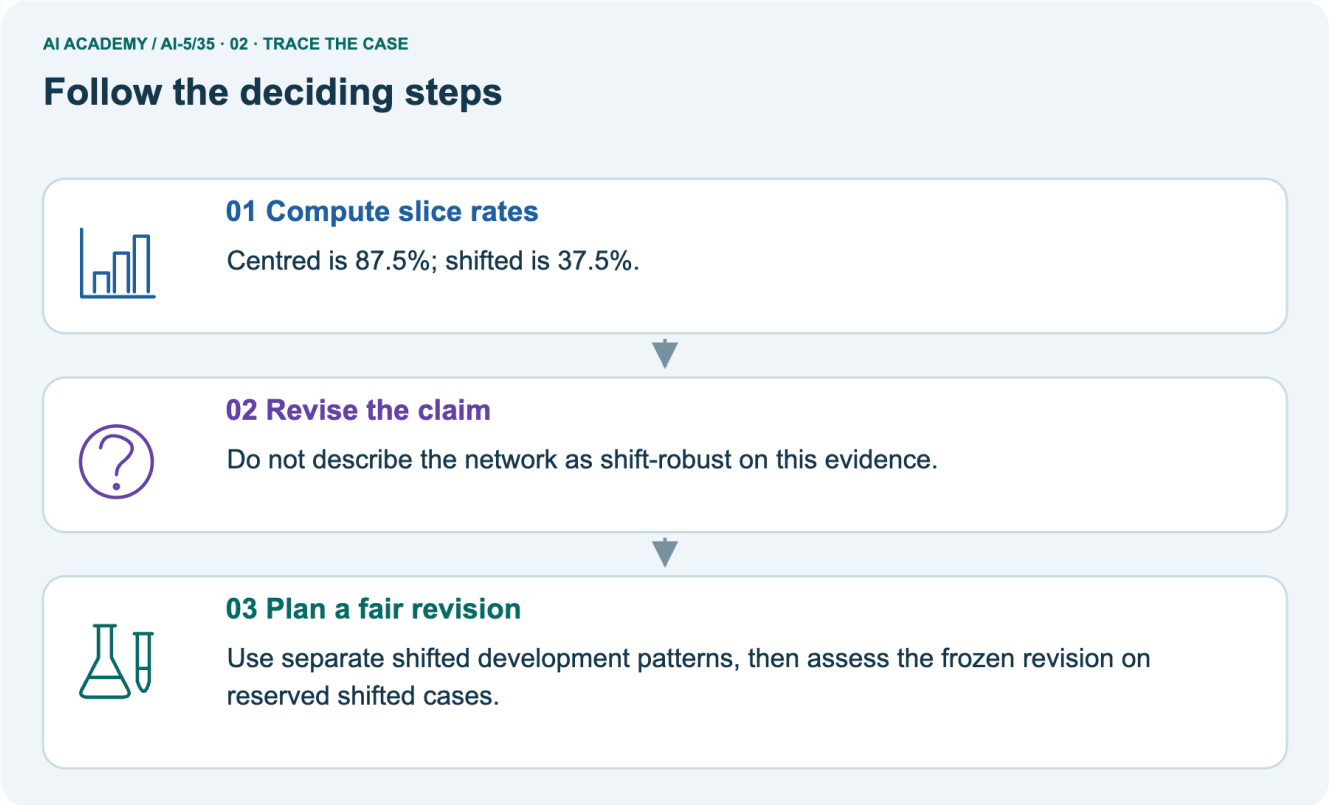
Follow the labelled picture with the matching explanation. Open the vector diagram

Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim. Supplied facts: Centred slice: 7/8 correct.

Shifted slice: 3/8 correct. The network and label rubric are frozen.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

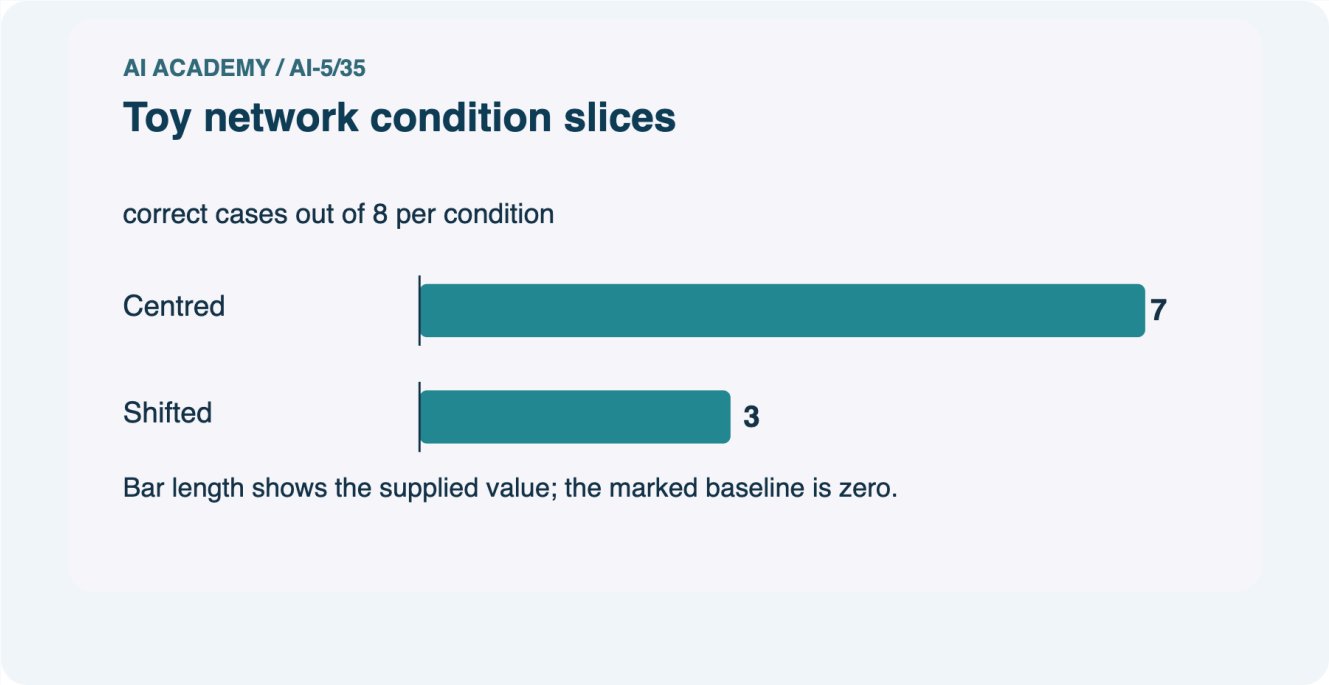


Follow the labelled picture with the matching explanation. Open the vector diagram

Compute slice rates: Centred is 87.5%; shifted is 37.5%. → Revise the claim: Do not describe the network as shift-robust on this **evidence**. → Plan a fair **revision**: Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Toy network condition slices

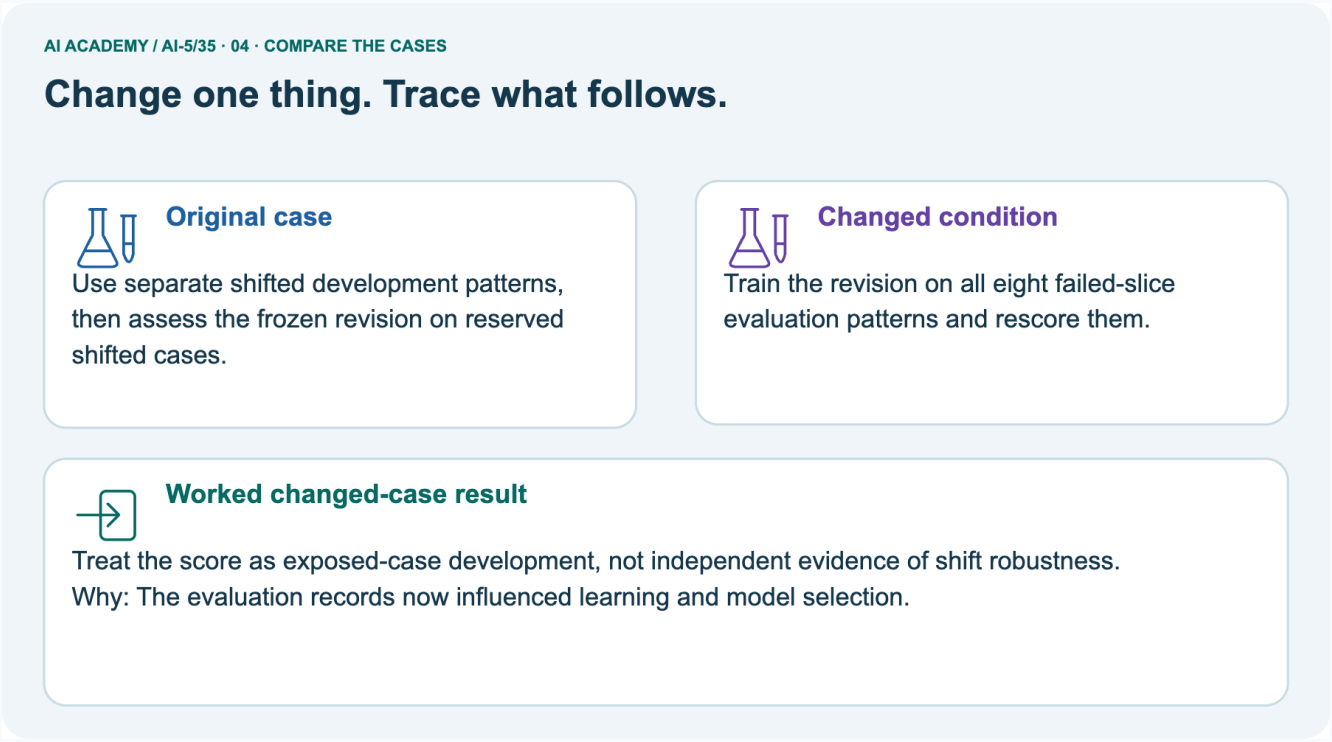


Follow the labelled picture with the matching explanation. Open the vector diagram

Toy network condition slices. Units: correct cases out of 8 per condition. Centred: 7. Shifted: 3.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



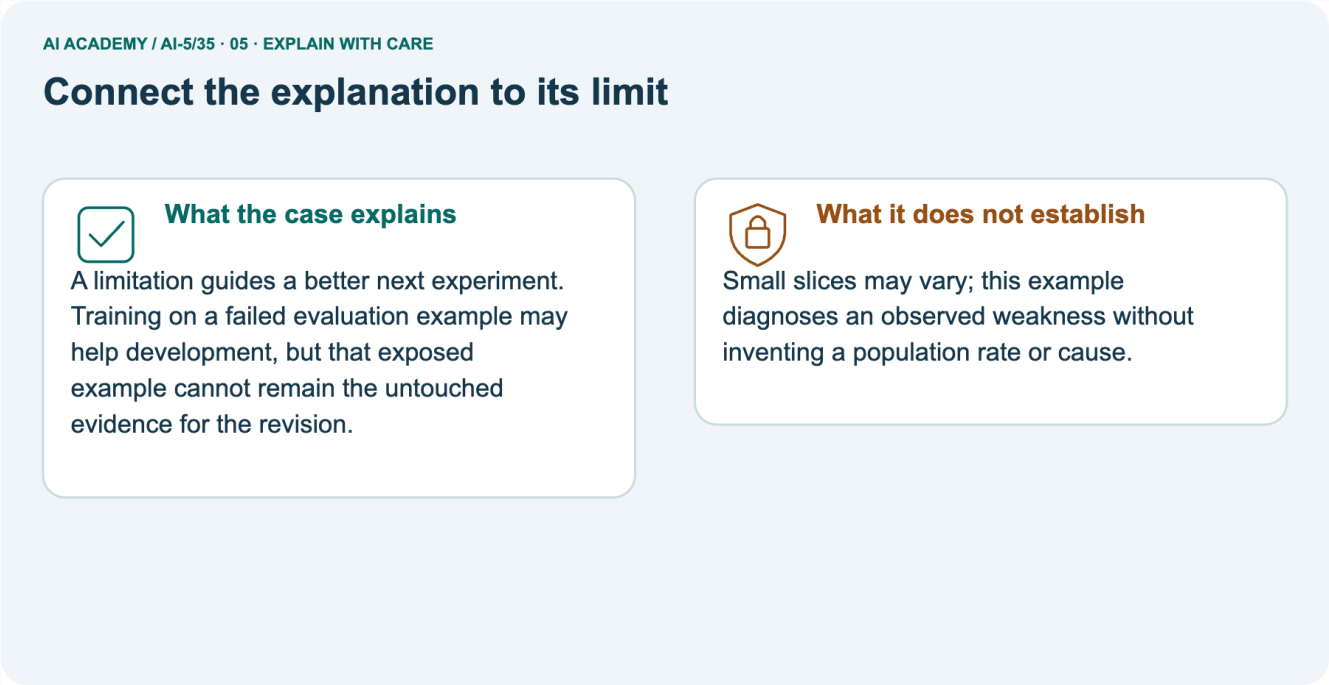
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases. Changed condition: Train the **revision** on all eight failed-slice evaluation patterns and rescore them.

Expected result: Treat the score as exposed-case development, not independent **evidence** of shift robustness. Reason: The evaluation records now influenced learning and model selection.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



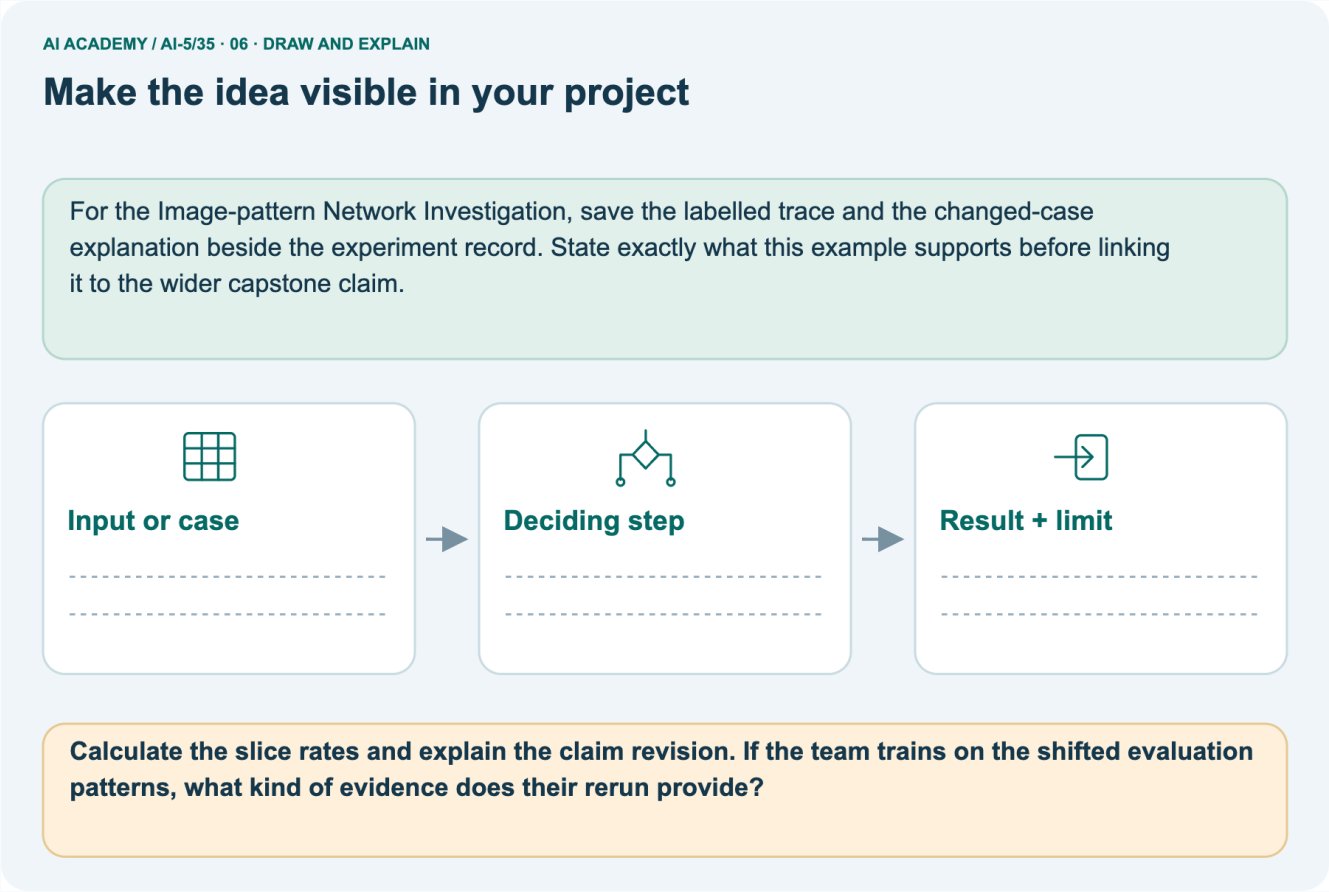
Follow the labelled picture with the matching explanation. Open the vector diagram

A limitation guides a better next experiment. **Training** on a failed evaluation example may help development, but that exposed example cannot remain the untouched **evidence** for the **revision**.

Limit: Small slices may vary; this example diagnoses an observed weakness without inventing a population rate or cause.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the slice rates and explain the claim **revision**.

If the team trains on the shifted evaluation patterns, what kind of **evidence** does their rerun provide?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.